

Market- versus Tax-Based Information: Effects on Gas-Bill Expectations and Demand for Residential Energy Efficiency

Pre-Analysis Plan*

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Abstract

Do households react more strongly to a gas-bill increase when it is attributed to a tax rather than to market forces? We address this question through a randomized information experiment with a representative sample of 1,500 Belgian adults. Participants will be assigned to a control group or to one of two treatments communicating the same expected increase in residential gas bills in 2028, attributed either to market forces or to the introduction of the European Emissions Trading System 2 (ETS 2). First, we will estimate whether tax-based information has stronger effects on longer-term gas-bill expectations than market-based information and examine whether responses vary with education and prior uncertainty. Second, we will investigate whether experimentally induced variation in long-term gas-bill expectations affects preferences for energy efficiency in housing, beliefs about aggregate renovation activity, and individual renovation intentions. Finally, reduced-form estimates will capture the overall effects of the information treatments, including effects potentially operating through framing or salience.

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1 Motivation

Global warming is one of the major challenges of the 21st century. The European residential building sector accounts for approximately 20% of total greenhouse gas emissions (Directorate-General for Energy, 2023). In 2023, the European Commission endorsed the framework of the Emissions Trading System 2 (ETS 2), which will introduce a carbon pricing mechanism for the building and road transport sectors starting in 2028. Several empirical studies show that gasoline consumption responds much more strongly to tax-induced price changes than to market-driven ones, with the estimated tax elasticity being about two-and-a-half to four times larger than the price elasticity (e.g., Andersson, 2019; Li et al., 2014; Rivers & Schaufele, 2015). These authors attribute this differential response to two main mechanisms: salience and persistence. The salience mechanism suggests that tax changes are more visible and receive greater media coverage than comparable market-driven price fluctuations, prompting consumers to pay closer attention to energy costs and to adjust their behavior more strongly. The persistence mechanism posits that consumers may perceive tax increases as more permanent than temporary market shocks, thereby revising upward their long-term expectations of energy prices. As rational agents tend to adjust their gasoline consumption more in response to permanent rather than transitory price changes (Scott, 2012), this perceived permanence can further stimulate investment in energy-efficient technologies. The salience mechanism has been empirically validated. For example, Li et al. (2014) demonstrate that, within the context of gasoline taxes, highly visible price components elicit stronger behavioral responses. Similarly, Basaglia et al. (2023) show that greater public and media attention to fuel taxes heightens their perceived cost, thereby intensifying demand reactions. In contrast, empirical evidence on the persistence mechanism remains relatively limited in the energy sector. Quantifying this mechanism requires data on individuals’ long-term energy price expectations, particularly when respondents are exposed to various types of price increases. While Anderson et al. (2013) find no evidence supporting persistence—using survey data on long-term price expectations—they implicitly assume respondents are fully aware of the tax component embedded in the gasoline price, which is unlikely in practice. Conversely, Li et al. (2014) provide suggestive evidence of persistence by estimating autocorrelation coefficients of gasoline taxes and tax-exclusive prices across U.S. states, indicating that taxes are statistically more persistent than market prices. However, this approach relies on observed price dynamics rather than direct belief data, limiting its ability to fully capture consumers’ perceptions of the permanence of tax changes. This study aims to address this gap by using a randomized information experiment to examine how information about a short-term price shock influences individuals’ long-term energy price expectations,

depending on whether the shock is framed as stemming from a fiscal measure or not. In addition, we test the hypothesis that a fiscal shock may lead to stronger growth in long-term price expectations, as individuals might anticipate that taxes are likely to increase over time. Another key contribution is that we examine whether higher long-term energy price expectations translate into stronger intentions to invest in residential energy-efficiency improvements. While Alberini et al. (2013) find that individuals with higher expected energy prices are more likely to invest in energy efficiency, recent studies show that beliefs cannot be treated as exogenous in observational data¹. Analogously to the case of inflation expectations (Coibion et al., 2023), long-term energy price expectations are endogenous in observational data: households intending to renovate may convince themselves that future prices will rise (reverse causality), and unobserved traits such as pro-environmental attitudes could affect both beliefs and renovation intentions (omitted variables). Unlike them, we are able to use exogenously generated variation in energy price expectations from our randomly assigned information treatments to characterize how these expectations affect energy-efficient renovation intentions. To this end, we elicit individuals’ beliefs about future residential gas bills and their intentions to renovate their dwellings. We exploit the forthcoming introduction of ETS 2 to examine how different framings of the expected 2028 gas price increase affect individuals’ long-term residential gas-bill expectations (considering no changes in temperature and dwelling conditions). This exogenous variation in beliefs is then used to causally assess whether higher long-term energy bill expectations increase intentions to undertake energy-efficiency renovations in the residential building sector. Our identification strategy relies on the fact that most individuals are largely unaware of the upcoming implementation of ETS 2 in 2028², although it is expected to have a substantial impact on household energy bills.

¹ In Coibion et al. (2023), the authors emphasize that observational correlations between inflation expectations and household spending cannot be interpreted causally because two fundamental sources of endogeneity are at play. First, reverse causality arises when households who plan to increase their spending revise their inflation expectations upward to rationalize their intended behavior. Second, omitted-variable bias occurs because unobserved shocks—such as experiencing unexpected increases in local prices—simultaneously raise households’ expenditures and their inflation expectations.

² See, for instance, the report by Bernard et al. (2025).

2 Experimental Design³

2.1 Pre-treatment questions

2.1.1 Survey entry and randomization

After choosing a language and providing informed consent, respondents are randomly assigned by LimeSurvey, with equal probability, to the control group, Treatment 1, or Treatment 2.

2.1.2 Prior gas-bill beliefs

Before treatment, respondents first provide point estimates of the average Belgian household's monthly natural-gas heating bill in 2028, 2030, and 2035. The question states that the corresponding household is expected to pay a monthly gas bill of X euros in 2027 and asks respondents to hold the household's building conditions and heating temperature constant. For each forecast year, respondents then report their confidence on a scale from 1 (not confident at all) to 7 (very confident).

2.1.3 Housing characteristics and prior renovation intentions

Respondents subsequently answer questions about their dwelling and baseline renovation intentions. These cover housing tenure, heating source, Energy Performance Certificate, dwelling type, construction period, and the number of bedrooms and office/study rooms. Owners are also asked which energy renovations they would be willing to undertake during the next ten years and how much they plan to invest in the selected works.

2.1.4 Socio-demographic characteristics

Finally, respondents report their gender, age, region, household composition, income, education, and employment status. The income wording distinguishes respondents who live with a partner from those who do not.

2.2 Treatment

2.2.1 Control group

Respondents in the control group receive no treatment information and no awareness question. They proceed directly to the common demonstration of the probability-distribution tool described below.

³The full questionnaire is provided in a separate document.

2.2.2 Treatment 1: Information about a 2028 gas-bill increase due to market forces

Respondents assigned to Treatment 1 see the following communication:

*In 2027, the average Belgian household heating with natural gas is expected to pay a monthly gas bill of X euros. Analysts expect this bill to increase by 18 euros in 2028, as gas prices are expected to rise due to supply and demand mechanisms.*⁴

They then answer the mandatory yes/no question: “Did you know that analysts expect gas prices to rise in 2028?”

2.2.3 Treatment 2: Information about a 2028 gas-bill increase due to a gas tax

Respondents assigned to Treatment 2 see the following communication:

In 2027, the average Belgian household heating with natural gas is expected to pay a monthly gas bill of X euros. Analysts expect this bill to increase by 18 euros in 2028 due to the introduction of a Tax on Gas by the European Commission.

They then answer the mandatory yes/no question: “Did you know that the European Commission will implement a tax on gas in 2028?”

2.2.4 Common demonstration

After the treatment page, all three groups see the same animated demonstration of the subjective probability-distribution tool. It shows how to drag selected probability bars, how the total changes as probabilities are entered, and how the *Normalize* button rescales the distribution to 100% and displays its average. Respondents must check “OK, I understand” before continuing.

2.3 Post-treatment questions

2.3.1 Outcomes (1): Residential Gas-Bill Expectations

We relicit Belgian residential gas-bill expectations for 2028, 2030, and 2035 on three successive pages using an interactive subjective probability-distribution interface. Respondents allocate probabilities across 23 displayed categories: less than €30, the interior bill levels from €40 to €240 in €10 increments, and more than €250. At least one category must receive a positive probability. The *Normalize* button rescales the entered probabilities to 100%; if respondents

⁴ The monthly increase of €18 is calibrated from a natural-gas price increase of €12 per MWh, based on a carbon intensity of 0.2 tCO₂ per MWh of natural gas (Intergovernmental Panel On Climate Change, 2006) and a €60 per tCO₂ ETS 2 carbon price. This price reflects an inflation adjustment of the initial €45 per tCO₂ level defined in 2020, consistent with Federal Planning Bureau (2025).

proceed without clicking it, the interface normalizes the distribution automatically before submission. The total is displayed in red whenever it differs from 100% and in green when it equals 100%. The respondent’s expected average monthly bill is displayed only for a normalized distribution and disappears if the bars are subsequently changed. On the 2030 and 2035 pages, the normalized averages from the preceding forecast years are also displayed.⁵

2.3.2 Outcomes (2): Aggregate energy-efficient renovation beliefs

Respondents are informed that the annual energy-renovation rate in Belgium was around 1% per year during the preceding ten years. They then provide a percentage estimate of the annual renovation rate during the next ten years.

2.3.3 Outcomes (3): Housing preferences

Respondents imagine that they are looking for a new place to live and rate the importance of five attributes on a scale from 0 (not important at all) to 10 (extremely important): location, dwelling size, energy performance, garden/outdoor space, and aesthetic appeal.

2.3.4 Outcomes (4): Energy-efficient renovation intentions

To measure renovation intentions, respondents are presented with a hypothetical budget of €50,000 to invest in renovating their dwelling and are asked to select the works they would undertake. The list includes both energy-efficiency and non-energy-efficiency improvements. Each option is assigned a fixed average cost estimate, which is deducted from the remaining budget when selected. Respondents cannot select a combination costing more than €50,000, and any unspent amount is retained as savings.

2.3.5 Financial literacy measure

The final question asks respondents how much €100 left for two years in a savings account paying 2% interest per year would be worth. This measure is collected after all post-treatment outcomes.

3 Setting and Sample Size

The sample consists of 1500 respondents and is constructed to be representative of the Belgian adult population with respect to age, gender, and region. Participants are randomly allocated

⁵ In line with Coibion et al. (2022), we vary the formulation of the second elicitation to minimize potential biases arising from survey fatigue and experimenter demand effects.

to three equal-sized groups.

3.1 Power Analysis

To have 80 percent power to detect an effect size of 0,2 of a standard deviation⁶ at a 5 percent significance level, we need 400 subjects in each treatment cell, or 1200 subjects in total. It is worth noting that in our main specifications, we will include repeated measurement, which decreases substantially the needed sample size for a given power and significance level (Clifford et al., 2021). However, we slightly increase the required sample size as we include interaction effects with a partially attenuated interaction, which are typically smaller than main effects (Sommet et al., 2023).

4 Main hypotheses and Mechanisms

Hypothesis 1: We expect the first and second treatment to have a similar positive impact on residential gas-bill expectations for 2028, and the second treatment to have a larger impact than the first treatment for gas-bill expectations of 2030 and 2035.

Hypothesis 2: As noted by Li et al. (2014), we expect heterogeneity in responses to the carbon tax treatment: more educated individuals are more likely to understand that ETS 2 is a permanent policy and that the carbon tax is expected to continue increasing over time, whereas others may not make this connection. Following Roth and Wohlfart (2020), we also expect to observe Bayesian updating, whereby individuals with greater uncertainty place more weight on the information provided by the treatment than those who are more confident in their prior beliefs.

Hypothesis 3: We expect that individuals whose residential gas-bill expectations are influenced by the treatment will foresee a higher annual rate of energy-efficient renovations in Belgium, exhibit stronger preferences for energy efficiency, and show a higher propensity to undertake renovations in their own homes.

5 Analysis

5.1 Baseline Balance

We will test for baseline balance between the different groups for the following variables:

⁶ A pilot study has revealed that the standard deviation of long-term residential gas price expectations was around 30, and the minimum effect that we want to detect is 6. In terms of energy bills it means that we want to detect an effect of 9 euro.

- Gender
- Age
- Income
- Education
- Employment status

We will regress each of these variables on a treatment indicator to see if there are imbalances.

5.2 Residential Gas-Bill Expectations

5.2.1 General Treatment Effect on Gas-Bill Expectations

We first assess the general effect of both treatments on individuals' (2028, 2030 and 2035) residential gas-bill expectations.

$$B_{i,t} = \alpha_t + \alpha_{1,t} T1_i + \alpha_{2,t} T2_i + \alpha_{3,t} B_{i,t}^{\text{prior}} + \Phi_t' \mathbf{X}_i + \varepsilon_{i,t} \quad (5.2.1)$$

- $B_{i,t}$: individual i 's gas bill expectation for year t ($t=2028, 2030$ or 2035)
- $T1_i$: A binary variable equal to 1 if individual i is assigned to the first treatment (price increase without further information), 0 otherwise
- $T2_i$: A binary variable equal to 1 if individual i is assigned to the second treatment (price increase due to ETS 2), 0 otherwise
- $B_{i,t}^{\text{prior}}$: pre-treatment gas bill expectation for year t ($t=2028, 2030$ or 2035)
- $\mathbf{X}_i$: Vector of individual-level control variables (gender, income, education).

5.2.2 Heterogeneous Treatment Effects on Gas-Bill Expectations

Second, we aim to assess whether certain individuals are more inclined to incorporate the information about ETS 2 into their long-term expectations than others.

$$B_{i,t} = \alpha_t + \alpha_{1,t} T1_i + \alpha_{2,t} T2_i + \sigma_t' \mathbf{S}_{i,t} + \delta_t' (T2_i \times \mathbf{S}_{i,t}) + \alpha_{3,t} B_{i,t}^{\text{prior}} + \varepsilon_{i,t} \quad (5.2.2)$$

- $\mathbf{S}_{i,t}$ contains education and pre-treatment uncertainty for forecast year t . Pre-treatment uncertainty is defined as $U_{i,t}^{\text{prior}} = 8 - C_{i,t}$, where $C_{i,t}$ is individual i 's confidence in their

pre-treatment expectation for year t , measured on the original scale from 1 (not confident at all) to 7 (very confident). Thus, $U_{i,t}^{\text{prior}}$ ranges from 1 to 7, with higher values indicating greater uncertainty.

5.3 Behavioral effects

5.3.1 Impact of Gas-Bill Expectations on Beliefs about Global Energy-Efficiency Renovation and Housing Preferences

Here, we examine the effect of long-term gas-bill expectations on beliefs about the overall level of energy-efficient renovations in Belgium and on housing preferences. These outcomes are continuous. Estimating their relationship with bill expectations using simple regressions would raise identification concerns due to endogeneity (see 1). To address these issues, we instrument long-term gas-bill expectations. We estimate the following equation, for which we interpret the coefficient $\beta_{1,t}$:

$$Y_i = \beta_t + \beta_{1,t} \hat{B}_{i,t} + \beta_{2,t} B_{i,t}^{\text{prior}} + \phi_t' \mathbf{X}_i + \mu_{i,t} \quad (5.3.1)$$

- Y_i : Continuous outcome measuring either (i) beliefs about the overall importance of global renovation in Belgium, or (ii) beliefs about housing preferences.
- $\hat{B}_{i,t}$: Instrumented post-treatment gas-bill expectations for $t \in \{2030, 2035\}$.

– We use the following first stage:

$$B_{i,t} = \alpha_t + \alpha_{1,t} T1_i + \alpha_{2,t} T2_i + \alpha_{3,t} B_{i,t}^{\text{prior}} + \theta_t' \mathbf{X}_i + \varepsilon_{i,t} \quad (5.3.2)$$

- $\mathbf{X}_i$: Vector of individual-level control variables (Age, Household, Income, Gender, Region, Family size, Employment status).

The causal interpretation of the IV estimates relies on the exclusion restriction that treatment assignment affects housing preferences, aggregate renovation beliefs, and individual renovation intentions only through its effect on gas-bill expectations. This assumption may be violated if the treatments directly affect these outcomes through policy salience, environmental concerns, or perceptions of the European Union. As in Coibion et al. (2024), we acknowledge this limitation when interpreting the IV estimates.

5.3.2 Impact of Gas-Bill Expectations on Investment Intentions in Energy Efficiency

We examine the causal impact of gas-bill expectations on individuals' intentions to invest in energy-efficiency improvements in their dwelling over the next ten years. Baseline renovation intentions are included as controls to improve precision. For this part of the analysis, we exclude individuals with no margin for renovation—such as renters, people living in apartments, and owners of well-insulated homes.

$$I_i^{EE} = \beta_t + \beta_{1,t} \hat{B}_{i,t} + \beta_{2,t} B_{i,t}^{\text{prior}} + \beta_{3,t} I_i^{EE,\text{prior}} + \phi_t' \mathbf{X}_i + \mu_{i,t} \quad (5.3.3)$$

- $\hat{B}_{i,t}$: Instrumented post-treatment gas-bill expectations for $t \in \{2030, 2035\}$.

– We use the following first stage:

$$B_{i,t} = \alpha_t + \alpha_{1,t} T1_i + \alpha_{2,t} T2_i + \alpha_{3,t} B_{i,t}^{\text{prior}} + \alpha_{4,t} I_i^{EE,\text{prior}} + \theta_t' \mathbf{X}_i + \varepsilon_{i,t} \quad (5.3.4)$$

- I_i^{EE} : an investment-intention measure capturing either (i) the number of energy-efficiency renovations individual i intends to undertake, or (ii) the amount (in euros) they plan to invest in energy efficiency.
- $I_i^{EE,\text{prior}}$: Pre-treatment (i) number of renovations intended by individual i and (ii) amount (in euros) they plan to invest in energy efficiency.

5.3.3 Reduced-form effects on housing preferences and renovation outcomes

We also investigate whether the treatments affect behavior in ways that cannot be explained solely by their impact on gas-bill expectations. We estimate the reduced-form effects of the two information treatments on (i) housing preferences, and (ii) beliefs about the aggregate energy-efficiency renovation rate in Belgium. Differences in renovation outcomes that are not accompanied by corresponding differences in gas-bill expectations would be consistent with a direct framing or salience channel, although they would not uniquely identify salience.

$$Y_i = \gamma + \gamma_1 T1_i + \gamma_2 T2_i + \mathbf{\Gamma}' \mathbf{X}_i + \varepsilon_i, \quad (5.3.5)$$

- Y_i : Outcome measuring (i) housing preferences, (ii) beliefs about the aggregate energy-efficiency renovation rate in Belgium, and (iii) individual intentions to invest in energy-efficiency renovations.

Finally, we measure the impact of the treatments on individual intentions to invest in energy-efficiency renovations.

$$Y_i = \gamma + \gamma_1 T1_i + \gamma_2 T2_i + \gamma_3 Y_i^{\text{prior}} + \mathbf{\Gamma}' \mathbf{X}_i + \varepsilon_i, \quad (5.3.6)$$

- Y_i : Outcome measuring individual intentions to invest in energy-efficiency renovations.
- Y_i^{prior} : Pre-treatment control for renovation intentions.

5.4 Multiple Hypothesis adjustment

We will report both unadjusted p-values and q-values controlling the False Discovery Rate (Benjamini et al., 2006) across these five outcomes at the 5% level.

6 Definition of Variables

6.1 Main outcomes

Our outcome variables are the following:

- Post-treatment 2028, 2030 and 2035 residential gas-bill expectations : Coded as continuous (see Appendix A for the construction of the variable)
- Renovation intentions (walls, roof, floor, windows, heat pump, solar panels) for the next 10 years: Coded as continuous
- Amount (in euro) for renovation intentions: Coded as continuous
- Annual Renovation percentage beliefs for Belgium in the next 10 years: Coded as continuous
- Relative importance of energy-efficiency in housing: Coded as continuous

6.2 Control variables

When the background questions are used as controls or interactions in the regressions, they will be coded as follows:

- Pre-treatment residential gas bill expectation for 2028, 2030 and 2035 : Coded as continuous
- Pre-treatment uncertainty for 2028, 2030 and 2035: Coded as $U_{i,t}^{\text{prior}} = 8 - C_{i,t}$ on a continuous scale from 1 to 7, separately for each forecast year

- Number of Pre-treatment renovation intentions : Coded as continuous
- Amount (in euro) for pre-treatment renovation intentions : Coded as continuous
- Education : coded as categorical variable
- Age : coded as continuous
- Household Income: coded as continuous
- Gender: coded as dummy
- Region : coded as categorical variable
- Family size: coded as continuous
- Employment status: coded as categorical variable

References

- Alberini, A., Banfi, S., & Ramseier, C. (2013). Energy efficiency investments in the home: Swiss homeowners and expectations about future energy prices. *The Energy Journal*, 34(1), 49–86. <https://doi.org/10.5547/01956574.34.1.3>
- Anderson, S. T., Kellogg, R., & Sallee, J. M. (2013). What do consumers believe about future gasoline prices? *Journal of Environmental Economics and Management*, 66(3), 383–403. <https://doi.org/10.1016/j.jeem.2013.07.002>
- Andersson, J. J. (2019). Carbon taxes and CO2 emissions: Sweden as a case study. *American Economic Journal: Economic Policy*, 11(4), 1–30. <https://doi.org/10.1257/pol.20170144>
- Basaglia, P., Behr, S. M., & Drupp, M. A. (2023). *De-fueling externalities: How tax salience and fuel substitution mediate climate and health benefits* (Working Paper No. 2041). DIW Discussion Papers. Retrieved October 31, 2025, from <https://www.econstor.eu/handle/10419/300888>
- Benjamini, Y., Krieger, A. M., & Yekutieli, D. (2006). Adaptive linear step-up procedures that control the false discovery rate. *Biometrika*, 93(3), 491–507. <https://doi.org/10.1093/biomet/93.3.491>
- Bernard, H., de Crevoisier, L., & Laine, N. (2025, October). *Transition des mobilités : anatomie d’une impasse*. Institut Montaigne. Retrieved October 30, 2025, from <https://www.institutmontaigne.org/publications/transition-des-mobilites-anatomie-dune-impasse>
- Clifford, S., Sheagley, G., & Piston, S. (2021). Increasing precision without altering treatment effects: Repeated measures designs in survey experiments. *American Political Science Review*, 115(3), 1048–1065. <https://doi.org/10.1017/S0003055421000241>
- Coibion, O., Georgarakos, D., Gorodnichenko, Y., Kenny, G., & Weber, M. (2024). The effect of macroeconomic uncertainty on household spending. *American Economic Review*, 114(3), 645–677. <https://doi.org/10.1257/aer.20221167>
- Coibion, O., Georgarakos, D., Gorodnichenko, Y., & van Rooij, M. (2023). How does consumption respond to news about inflation? field evidence from a randomized control trial. *American Economic Journal: Macroeconomics*, 15(3), 109–152. <https://doi.org/10.1257/mac.20200445>

- Coibion, O., Gorodnichenko, Y., & Weber, M. (2022). Monetary policy communications and their effects on household inflation expectations. *Journal of Political Economy*, 130(6), 1537–1584. <https://doi.org/10.1086/718982>
- Directorate-General for Energy. (2023). EU energy in figures: Statistical pocketbook 2023. Retrieved July 9, 2024, from <https://data.europa.eu/doi/10.2833/502436>
- Federal Planning Bureau. (2025). *Exploration de l'impact de l'ETS2 sur les dépenses de chauffage des ménages* (21).
- Intergovernmental Panel On Climate Change. (2006). *IPCC guidelines for greenhouse gas inventories* (Vol. 2). IGES. <https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html>
- Li, S., Linn, J., & Muehlegger, E. (2014). Gasoline taxes and consumer behavior. *American Economic Journal: Economic Policy*, 6(4), 302–342. <https://doi.org/10.1257/pol.6.4.302>
- Rivers, N., & Schaufele, B. (2015). Salience of carbon taxes in the gasoline market. *Journal of Environmental Economics and management*, 74, 23–36.
- Roth, C., & Wohlfart, J. (2020). How do expectations about the macroeconomy affect personal expectations and behavior? *The Review of Economics and Statistics*, 102(4), 731–748. https://doi.org/10.1162/rest_a_00867
- Scott, K. (2012). Rational habits in gasoline demand. *Energy Economics*, 34(5), 1713–1723. <https://doi.org/10.1016/j.eneco.2012.02.007>
- Sommet, N., Weissman, D. L., Cheutin, N., & Elliot, A. J. (2023). How many participants do i need to test an interaction? conducting an appropriate power analysis and achieving sufficient power to detect an interaction. *Advances in Methods and Practices in Psychological Science*, 6(3). <https://doi.org/10.1177/25152459231178728>

A Variable construction

- Post-treatment expectations of residential natural gas bills for the years 2028, 2030, and 2035 are elicited using an interactive subjective probability distribution. For each forecast year, respondents assign probabilities to 23 displayed categories: less than €30, the interior bill levels from €40 to €240 in €10 increments, and more than €250. The interface normalizes the probabilities to sum to one. We summarize these beliefs by computing the subjective expected value of the distribution:

$$B_{i,t} = \sum_{k \in \mathcal{K}} p_{i,t,k} x_k \quad (\text{A.0.1})$$

where $B_{i,t}$ denotes the expected monthly residential gas bill anticipated by individual i for forecast year t ; $\mathcal{K} = \{< 30, 40, \dots, 240, > 250\}$ is the set of displayed categories; $p_{i,t,k}$ is the subjective probability assigned by individual i to category k for year t , with $\sum_{k \in \mathcal{K}} p_{i,t,k} = 1$; and x_k is the numerical value used to summarize category k . The open-ended endpoint categories are bottom- and top-coded at €30 and €250, respectively, so that $x_k \in \{30, 40, \dots, 250\}$.